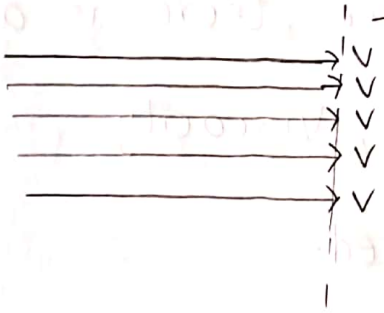
Viscosity

* The internal friction developed b/w layers of the fluid that opposes the relative motion b/w them.

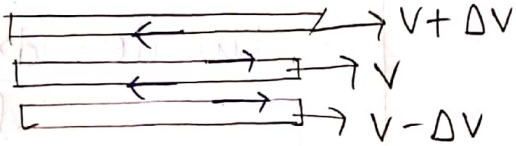
*



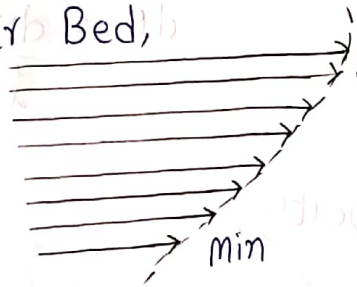
→ Velocity Profile [Straight Line].

⇒ Ideal Fluid (non-viscous fluid)

* Real Fluid (viscous fluid)

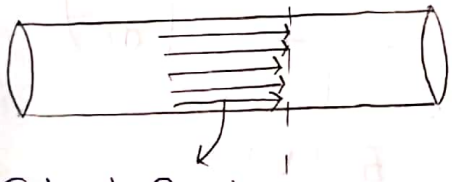


For River Bed,



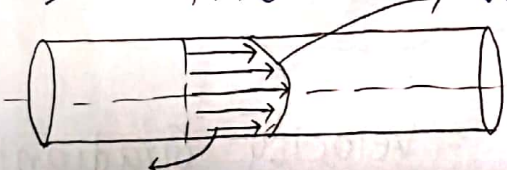
→ Velocity Profile [Parabola]

* Capillary Tube,



→ Velocity Profile [Straight Line]

Ideal Fluid.



→ Velocity Profile

[~~Circle~~ Parabola]

Real Fluid

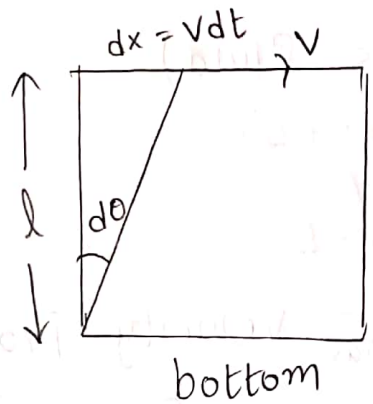
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* Velocity is minimum at walls of Tube & maximum at Axis of Tube [middle]

* As there is change in momentum b/w layers per unit time, there is a internal force b/w them i.e, Viscocity

⇒ Heat is produced.

Coefficient of Viscocity :-



$$\tan d\theta = \frac{dx}{l}$$

$$d\theta = \frac{dx}{l}$$

Coefficient of Viscocity,

$$\eta = \frac{\text{Shearing Stress}}{\text{Strain Rate}} = \frac{\frac{F}{A}}{\left(\frac{d\theta}{dt}\right)}$$

$$\eta = \frac{F}{\frac{A \cdot \left(\frac{dx}{dt}\right)}{l}} = \boxed{\frac{F}{A \cdot \left(\frac{v}{l}\right)}}$$

$$\frac{v}{l} = \frac{v-0}{l-0} = \frac{dv}{dy} = \text{Velocity Gradiant}$$

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* Velocity gradiant is ⊥ to fluid flow

Units of Velocity Gradiant,

$$\frac{dv}{dy} \rightarrow \frac{m}{s \times m} = s^{-1}$$

* $$\eta = \frac{F}{A \cdot \left(\frac{dv}{dy}\right)}$$

Units of η :-

$$\eta = \frac{N}{m^2 s^{-1}} = Ns/m^2 \text{ (or) Poisuellie } (P_l)$$

$$C \cdot G \cdot S = \text{Dyne } s/cm^2 \text{ (or) Poise}$$

$$1P_l = \frac{1N\text{-}s}{1m^2} = \frac{10^5 \text{ dyne-s}}{10^4 cm^2} = \frac{10 \text{ dyne-s}}{cm^2}$$

$$\boxed{1P_l = 10 \text{ poise}}$$

Effect of Temperature :-

* In case of liquids,

As Temperature ↑ ⇒ Cohessive forces ↓

⇒ Viscocity ↓

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* In case of Gases,

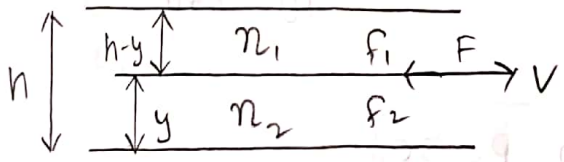
As Temperature ↑ ⇒ Collisions ↑

⇒ Viscocity ↑

Ex:- Through a very narrow gap of height 'h'. A thin plate of large extension is pulled at a velocity 'v' on one side of plate of oil of velocity η_1 & the other side oil of Viscocity η_2 ① Calculate the position of the plate so that shear force on the two sides of the plate is equal.

② The pull required to drag the plate is minimum is ?

Sol:-



① Let A = Area of plate

$$f_1 = \eta_1 A \left(\frac{v-0}{h-y}\right)$$

$$f_2 = \eta_2 A \left(\frac{v-0}{y}\right)$$

Given $f_1 = f_2$

$$\eta_1 A \left(\frac{v}{h-y}\right) = \eta_2 A \left(\frac{v}{y}\right)$$

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$$\frac{\eta_1}{h-y} = \frac{\eta_2}{y}$$

$$\eta_1 y = \eta_2 h - \eta_2 y$$

$$(\eta_1 + \eta_2) y = \eta_2 h$$

$$\boxed{y = \frac{\eta_2 h}{\eta_1 + \eta_2}}$$

(2) Plate is pulled with constant speed 'v'

F_{net} on plate $= 0$

$$F = f_1 + f_2$$

$$F = \frac{\eta_1 A V}{h-y} + \frac{\eta_2 A V}{y} \rightarrow (1)$$

For minimum pull,

$$\frac{dF}{dy} = 0$$

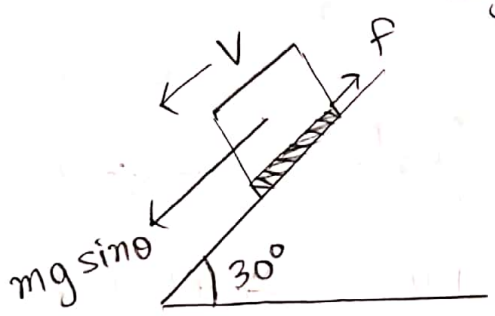
$$\Rightarrow \boxed{y = \frac{\sqrt{\eta_2} \cdot h}{\sqrt{\eta_1} + \sqrt{\eta_2}}}$$

Put y value in (1) to get minimum value

Ex:- A Cubical block of side 2m having mass 20kg slides with constant velocity of 10 m/s on an Inclined plane of 30°.

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lubricated with the oil of viscocity $\eta = 10^{-1}$ Poise. Take $g = 10\ m/s^2$. Calculate thickness of the layer of the liquid.



Sol:- $mg\sin\theta = f$ $[\because F_{net} = 0]$.

$$f = \eta \cdot A\left(\frac{V-0}{h}\right)$$

$$\Rightarrow mg\sin\theta = \eta \cdot A\left(\frac{V}{h}\right)$$

10^{-1} Poise $= 10^{-1} \times 10^{-1}\ P_l$

$= 10^{-2}\ P_l$.

$$20 \times 10 \times \frac{1}{2} = \frac{10^{-2} \times 4 \times 10}{h}$$

$$h = \frac{40 \times 10^{-2}}{10^2}$$

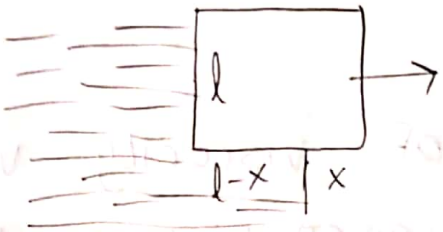
$$\boxed{h = 4\ mm}$$

Ex:- A massless metal plate is placed on a horizontal table top lubricated with oil. The sheet is a square of side length $l = 1m$ & oil layer has thickness $h = 1\ mm$. Initially, one edge of the sheet coincides with one edge of table. The sheet is pulled outwards without rotation with an

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constant force F = 50 is N. If coefficient of viscocity of oil is $\eta = 0.2\ Ns/m^2$. How long will it take to pull half of the sheet out of the table.

Sol:-



$$F = \eta A\left(\frac{dv}{dy}\right) \quad \Longleftarrow \quad F - f = ma^{\,0}$$

$$F = \eta\, l(l-x)\left(\frac{v-0}{h}\right)$$

$$V = \frac{Fh}{\eta\, l(l-x)}$$

$$\frac{dx}{dt} = \frac{Fh}{\eta\, l(l-x)}$$

$$\int_{x=0}^{x=l/2} (l-x)\,dx = \int_{t=0}^{t=t} \frac{Fh}{\eta l}\cdot dt$$

$$\left[lx - \frac{x^2}{2}\right]_0^{l/2} = \frac{Fht}{\eta l}$$

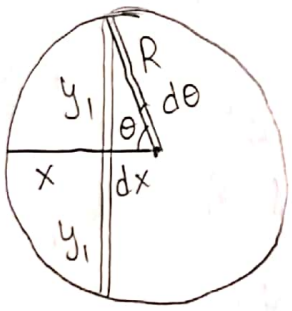
$$\frac{3l^2}{8} = \frac{Fht}{\eta l}$$

$$\Rightarrow \boxed{t = \frac{3l^3\eta}{8Fh}} \Rightarrow \boxed{t = 5\,sec}$$

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Ex:- Consider a disk of mass m, radius R lying on a liquid layer of thickness T & coefficient of viscocity η as shown in the fig. Answer the following quenstions :-

(1) The coefficient of viscocity varies as $\eta = \eta_0 x$ at the given instant the disk is floating towards right with a velocity V. Find the force required to move the disk slowly at the given instant.



Sol:- Since disc velocity is constant.

$\Rightarrow F_{net} = 0$

$\Rightarrow F_{viscous} = F$

From fig, $x = R - R\cos\theta$

$x = R(1-\cos\theta)$

$dx = R\sin\theta \cdot d\theta$

From fig, $\sin\theta = \frac{y_1}{R} \Rightarrow y_1 = R\sin\theta$

$y = 2y_1 = 2R\sin\theta$

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Area of elementary strip,

$$dA = (dx)\,y = (R\sin\theta\, d\theta)\, 2R\sin\theta$$

$$dA = 2R^2\sin^2\theta\, d\theta.$$

Viscous force at elementary strip,

$$dF = \eta\,(dA)\left(\frac{V-0}{T}\right)$$

$$dF = \frac{2\eta R^2\sin^2 d\theta \cdot V}{T}$$

Total force on the disc.

$$F = \int_{\theta=0}^{\theta=\pi} dF = \int_{\theta=0}^{\theta=\pi} \frac{2\eta_0 \times R^2\sin^2\theta\, V}{T}\, d\theta$$

$$F = \frac{2\eta_0 R^2 V}{T}\int_{\theta=0}^{\theta=\pi} R(1-\cos\theta)\cdot\sin^2\theta\, d\theta$$

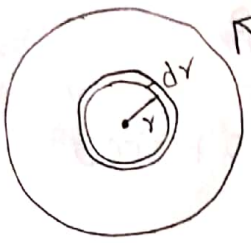
$$F = \frac{2\eta_0 R^3 V}{T}\int_{\theta=0}^{\theta=\pi} \sin^2\theta\,(1-\cos\theta)\cdot d\theta$$

$$F = \frac{2\eta_0 R^3 V}{T}\left[\int_0^{\pi}\sin^2\theta\cdot d\theta - \int_0^{\pi}\sin^2\theta\cdot\cos\theta\cdot d\theta\right]$$

$$\boxed{F = \frac{\pi\eta_0 R^3 V}{T}}$$

② The Torque required to rotate the disk at a constant angular velocity ω given the viscocity in uniformly η.

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$$\frac{dv}{dy} = \frac{v_2 - v_1}{y_2 - y_1} = \frac{r\omega - 0}{T}$$

Sol:- ω = constant

$\Rightarrow \tau_{net} = 0$

$\Rightarrow \tau_{ext} = \tau_{viscous}$

At the location of elementary strip,

Velocity of disc = $r\omega$

Area of elementary strip $(dA) = 2\pi r dr$

Viscous force,

$$dF = \eta (dA) \left(\frac{dv}{dy}\right)$$

$$dF = \eta (2\pi r dr) \left(\frac{r\omega}{T}\right)$$

Torque due to viscous force,

$$d\tau = (r)\, dF$$

$$d\tau = (r)\, \eta\, 2\pi r dr \left(\frac{r\omega}{T}\right)$$

$$d\tau = \frac{2\pi\eta\omega r^3 dr}{T}$$

Total Torque,

$$\tau = \int_{r=0}^{r=R} d\tau = \frac{2\pi\eta\omega}{T} \int_0^R r^3 dr = \frac{\pi\eta\omega R^4}{2T}$$

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(3) A disc rotating with angular velocity ω is placed on a viscous liquid of thickness T. Find the angle rotated by the disc before it comes to rest.

Sol:- Torque due to viscous force

$$\tau = \frac{\pi \eta \omega R^4}{2T} \quad \text{(Opposing Torque)}$$

$$\tau = I\alpha \quad ; \quad I = \frac{mR^2}{2}$$

$$\alpha = \frac{-d\omega}{dt} = -\omega \frac{d\omega}{d\theta}$$

$$\Rightarrow \frac{-mR^2}{2} \frac{d\omega}{dt} = \frac{\pi \eta \omega R^4}{2T}$$

$$\frac{-d\omega}{dt} = \frac{\pi \eta R^2 \omega}{mT}$$

$$\int_{\omega_0}^{\omega} \frac{-d\omega}{\omega} = \frac{\pi \eta R^2}{mT} \int_0^t dt$$

$$\ln\left(\frac{\omega_0}{\omega}\right) = \frac{\pi \eta R^2}{mT} t$$

$$\boxed{t = \frac{mT}{\pi \eta R^2} \ln\left(\frac{\omega_0}{\omega}\right)}$$

$$\tau = I\alpha$$

$$\frac{\pi \eta \omega R^4}{2T} = \frac{-mR^2}{2} \omega \cdot \frac{d\omega}{d\theta}$$

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$$-\int_{\omega_0}^{\omega} d\omega = \int_0^{\theta} \frac{\pi \eta R^2}{mT} d\theta$$

$$\omega_0 - \omega = \frac{\pi \eta R^2}{mT} \theta$$

$$\theta = \frac{(\omega_0 - \omega) mT}{\pi \eta R^2}$$

When disc stops, $\omega = 0$

$$\theta = \frac{\omega_0 mT}{\pi \eta R^2}$$

Total angle turned before coming to rest.

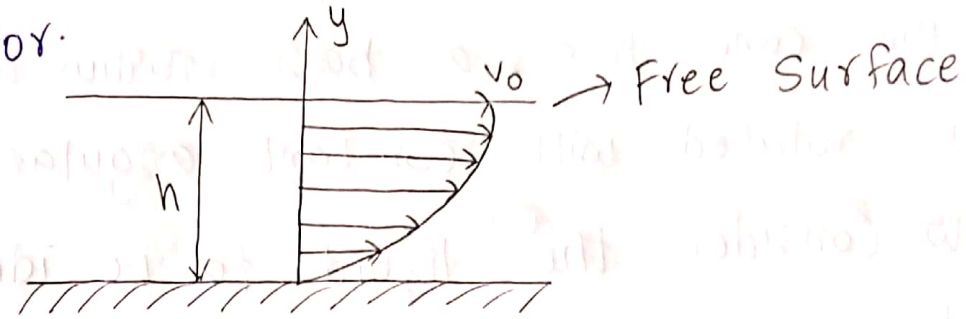
Number of rotations,

$$N = \frac{\theta}{2\pi}$$

Ex: A liquid is flowing through a horizontal channel. The speed of flow depends on height from the floor as $V = V_0 \left[2\left(\frac{y}{h}\right) - \left(\frac{y}{h}\right)^2\right]$. Where h is the height of liquid in the channel and V_0 is the speed of the top layer. Coefficient of

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viscosity is η. Calculate the shear stress that the liquid exerts on the floor.



Sol:- $V = V_0\left[2\left(\frac{y}{h}\right) - \left(\frac{y}{h}\right)^2\right]$

Velocity gradient $= \frac{dv}{dy} = V_0\left[\frac{2}{h} - 2\left(\frac{y}{h}\right)\times\frac{1}{h}\right]$

at bottom $y=0$, $\Rightarrow \frac{dv}{dy} = \frac{2V_0}{h}$

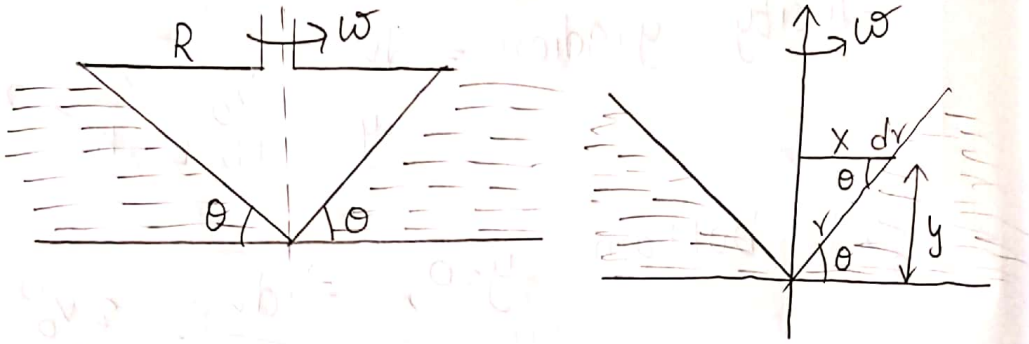
$\frac{F}{A} = \eta\left(\frac{dv}{dy}\right)$

Stress $= \frac{2\eta V_0}{h}$

Ex:- A viscometer, an instrument used to study characteristics of a non-ideal fluid, consists of a flag plate and a rotating cone. The cone has a large apex angle and the angle θ shown in figure is very small. The apex of the cone just

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touches the plate and a liquid fills the narrow gap blw the plate and the cone. The cone has a base radius R and is rotated with constant angular speed ω. Consider the liquid to be ideal and take its coefficient of viscocity to be η. Calculate the torque needed to drive the cone.



Sol:- From fig,

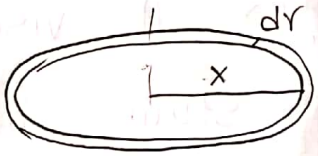
$$\cos\theta = \frac{x}{r} \Rightarrow r = \frac{x}{\cos\theta}.$$

$$\tan\theta = \frac{y}{x}$$

$$y = x\tan\theta$$

Here θ is constant

$$dr = \frac{dx}{\cos\theta}$$



dA = Area of elementary ring,

$$= 2\pi x\, dr = 2\pi x \frac{dx}{\cos\theta}$$

$$dF = \eta(dA)\left(\frac{dv}{dy}\right) = \eta\left(2\pi x \frac{dx}{\cos\theta}\right)\left(\frac{x\omega}{y}\right)$$

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$$dF = \frac{2\pi\eta\omega x^2 dx}{\cos\theta \cdot (x\tan\theta)} = \frac{2\pi\eta\omega x\, dx}{\sin\theta}$$

Torque required, $d\tau = (dF)\,x$

$$d\tau = \frac{2\pi\eta\omega x^2 dx}{\sin\theta}$$

Total Torque, $\tau = \int_{x=0}^{x=R} d\tau = \frac{2\pi\eta\omega}{\sin\theta} \int_{x=0}^{x=R} x^2 \cdot dx$

$$= \frac{2\pi\eta\omega}{\sin\theta}\left(\frac{R^3}{3}\right)$$

$$= \frac{2\pi\eta\omega R^3}{3\sin\theta}$$

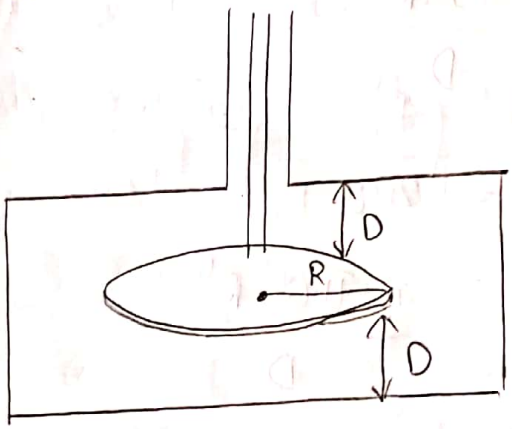
For small angles, $\sin\theta \approx \theta \Rightarrow$ $\boxed{T = \frac{2\pi\eta\omega R^3}{3\theta}}$

Ex:-

Find the power required to drive the disc at constant velocity. Given

$R = 10$ cm, $\omega = 600$ ~~rad~~ rev/min, $\eta = 8.5 \times 10^{-2}$ Pa-s

$D = 1.0$ mm.



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Sol:-

$$\omega = 600 \text{ rev/min} = \frac{600 \times 2\pi \text{ rad}}{60 \text{ sec}} = 20\pi \text{ rad/sec}$$

For a small elementary ring.

$$dF = \eta (dA) \left(\frac{dv}{dy}\right)$$

$$= \eta \cdot 2\pi r dr \left(\frac{r\omega - 0}{D}\right)$$

$$d\tau = r \cdot dF$$

$$d\tau = \eta \cdot 2\pi r^3 \frac{\omega}{D} \cdot dr$$

$$\tau = \int d\tau = \int_{r=0}^{r=R} \eta \cdot \frac{2\pi\omega}{D} r^3 \cdot dr$$

$$= \frac{\eta \cdot 2\pi\omega}{D} \left[\frac{R^4}{4}\right]$$

$$\tau = \frac{\eta \cdot \pi \omega R^4}{2D}$$

Total Torque $= 2\tau$

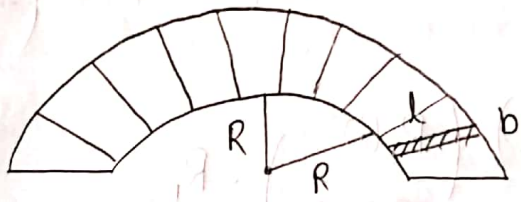
$$= \frac{\eta \cdot \pi \omega \cdot R^4}{D}$$

$$P = \tau \cdot \omega$$

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$$= \frac{\eta \cdot \pi \omega^2 R^4}{D}$$

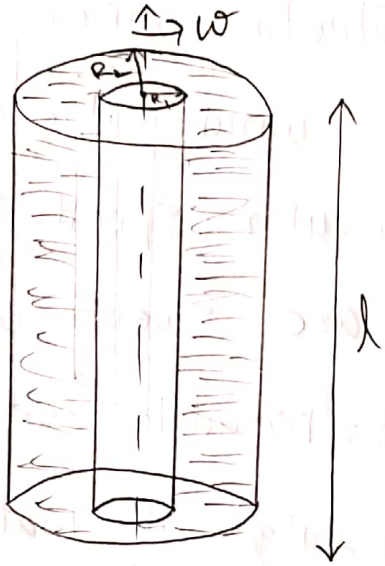
Ex:- Wind shield wiper plate sweeps the wet wind shield rotating at constant angular speed ω. R = Radius of inner most arc swept by the blade l = length, b = breadth of blade. η = Coefficient of viscocity of water, t = thickness of uniform layer of water on the glass surface. Calculate the Torque required to rotate the blade.



Sol:- $\frac{\eta b \omega}{3t} \left[(R+L)^3 - R^3\right] = \tau$

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Ex:- A liquid of viscocity η is filled b/w an outer fixed cylinder & inner cylinder. The Central Solid cylinder starts rotating with an Initial angular velocity ω_0. Calculate the time after which angular velocity of Central Cylinder become half of the initial velocity.



Sol:-

$$t = \frac{m(R_2 - R_1)\ln 2}{4\pi\eta l R_1}$$

$$F = \eta \cdot A \cdot \frac{dv}{dy} \Rightarrow dF = \eta \cdot (2\pi R_1 l)\left(\frac{R_1\omega - 0}{R_2 - R_1}\right)$$

$$d\tau = \eta \cdot (2\pi R_1 l)\left(\frac{R_1\omega}{R_2 - R_1}\right) \cdot R_1$$

$$\tau = I\alpha$$

$$\frac{\eta 2\pi R_1^3 l\omega}{R_2 - R_1} = \frac{m R_1^2}{2} \cdot \frac{d\omega}{dt}$$

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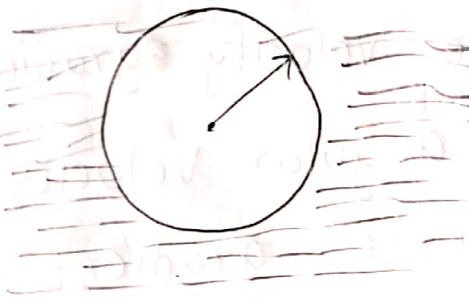
$$\int_0^t dt = \frac{m(R_2-R_1)}{4\pi\eta l R_1} \int_{\omega=\omega_0}^{\omega=\omega_0/2} \frac{d\omega}{\omega}$$

$$t = \frac{m(R_2-R_1)}{4\pi\eta l R_1} \left[\ln \frac{\omega_0}{\omega_0/2} \right]$$

$$\boxed{t = \frac{m(R_2-R_1)}{4\pi\eta l R_1} \cdot \ln 2}$$

Ex:- A Plastic disc of radius R = 10cm of oil film of thickness 2mm. Find torque required about its central vertical axis with constant angular velocity ω_0 ; $\omega = 8$ rad/sec , $\eta_{oil} = 1$ kg/m sec

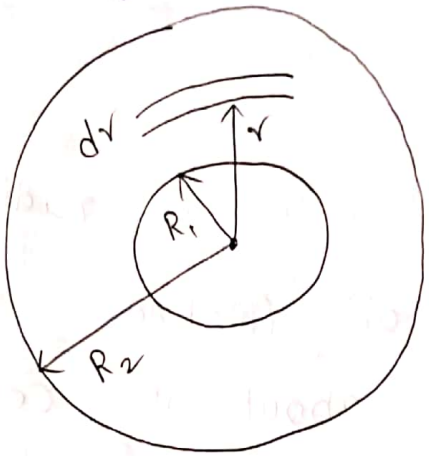
A:- 1 N/m



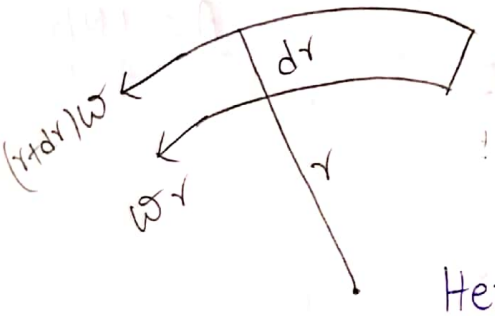
Sol:-

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IRODOV:- Outer Cylinder is rotating with ω.



① When there is no 'ω' gradient:-

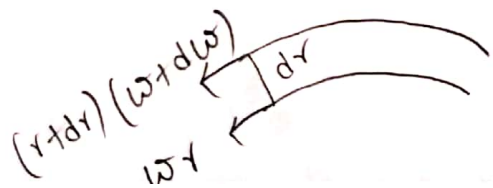


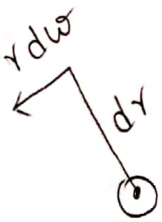
Here Velocity gradient ≠ 0 but angular Velocity gradient = 0

Here there is no shearing stain

⇒ Shearing Stress = 0

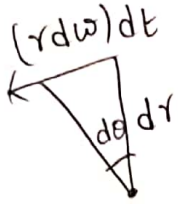
② With 'ω' gradient:-





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In the next dt-sec,



From fig,

$$\tan d\theta = \frac{(r d\omega) dt}{dr}$$

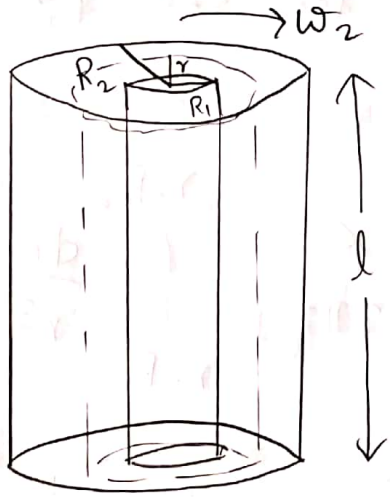
$$\frac{d\theta}{dt} = \frac{r d\omega}{dr}$$

$$\eta = \frac{\text{Stress } (\sigma)}{\text{Strain Rate}} = \frac{\sigma}{\left(\frac{d\theta}{dt}\right)}$$

$$\sigma = \eta \cdot \frac{d\theta}{dt}$$

$$\boxed{\sigma = \eta \cdot r \cdot \frac{d\omega}{dr}}$$

Ex:-



In Steady State, Torque at any intermediate layer = same

$$\sigma = \eta \cdot r \cdot \frac{d\omega}{dr} \rightarrow (1)$$

At a distance 'r'

$$\tau = \sigma A r = 2\pi r l r = c \text{ (constant)}$$

independent of radius.

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$$\sigma \cdot 2\pi r^2 l = C.$$

$$\sigma = \frac{C}{2\pi r^2 l}.$$

$$\eta \cdot r \cdot \frac{d\omega}{dr} = \frac{C}{2\pi r^2 l}.$$

$$\boxed{d\omega = \frac{C}{\eta \cdot 2\pi r^3 l} \cdot dr}$$

$$\omega = \int_0^{\omega} d\omega$$

$$\omega = \frac{C}{2\pi\eta \cdot l} \int_{r=R_1}^{r=r} \frac{dr}{r^3}$$

$$\omega = \frac{C}{2\pi\eta \cdot l} \cdot \frac{1}{2}\left[\frac{1}{R_1^2} - \frac{1}{r^2}\right] \rightarrow (1)$$

and also,

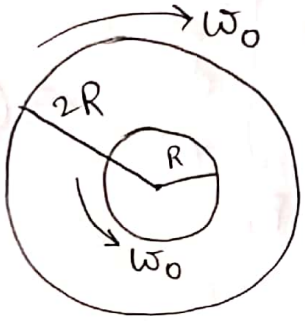
$$\int_0^{\omega_2} d\omega = \frac{C}{2\pi\eta \cdot l} \int_{r=R_1}^{r=R_2} \frac{dr}{r^3}.$$

$$\omega_2 = \frac{C}{2\pi\eta \cdot l} \cdot \frac{1}{2}\left[\frac{1}{R_1^2} - \frac{1}{R_2^2}\right] \rightarrow (2)$$

$$\frac{(1)}{(2)} \Rightarrow \boxed{\frac{\omega_1}{\omega_2} = \frac{(r^2 - R_1^2) R_2^2}{(R_2^2 - R_1^2) r^2}}$$

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Ex:-



At $r=r_0$, angular velocity of fluid elements = 0.

P = rate of heat generation in whole system per unit length of cylinder

τ = Torque

K·E = K·E per unit length of cylinder

(1) Mark the correct Statements

(A) τ at $r=\frac{4R}{3}$ = τ at $r=\frac{3R}{2}$

(B) The magnitude of angular velocity of gradient in radial direction of the fluid at a distance 'r' from axis is proportio-nal to $\frac{1}{r^3}$.

(C) $r_0 = \frac{3R}{2}$

(D) $r_0 = 2.\sqrt{\frac{2}{5}}\,R$.

Sol:- $$\int_{-\omega_0}^{\omega} d\omega = \frac{c}{2\pi\eta l}\int_{R_1}^{r}\frac{dr}{r^3} \rightarrow (1)$$

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$$\xi \int_{-\omega_0}^{\omega_0} d\omega = \frac{C}{2\pi\eta l} \int_{R_1}^{R_2} \frac{dr}{r^3} \rightarrow (2)$$

(2) Mark the incorrect Statements :-

(A) $P = \frac{32\pi\eta\omega_0^2 R^2}{3}$

(B) $P = \frac{64\pi\eta\omega_0^2 R^2}{3}$

(C) K.E is constant with time.

(D) K.E is decreasing with time.

Sol:- P = rate of work done by external agent

$$P = 2C\omega_0$$

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Critical Velocity (V_c) :-

* At low speeds, fluid speed is laminar.
* At higher speeds, fluid flow is turbulant
* Maximum speed of the fluid particle at which flow changes from laminar to turbulant is called Critical Velocity.



1. $V_c \propto \eta$ (coefficient of viscocity)
2. $V_c \propto \frac{1}{\rho}$ (Density of fluid)
3. $V_c \propto \frac{1}{D}$ (diameter of tube of flow)

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On Combining,

$$V_c \propto \frac{\eta}{\rho D}$$

$$\boxed{V_c = \frac{R\eta}{\rho D}}$$

R = Reynold's number

R has no units & dimensions.

$$\boxed{R = \frac{\rho D V_c}{\eta}}$$

Points:-

1. $R < 2000$, flow is laminar
2. $R > 3000$, flow is turbulant
3. $2000 < R < 3000$, flow is unsteady.

Coefficient of Dynamic Viscocity / Kinematic Viscocity (μ):-

$$\boxed{\mu = \frac{\eta}{\rho}}$$

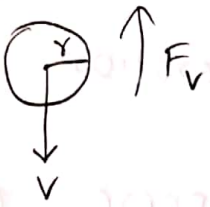
Units:- $\frac{Pa\text{-}s}{\frac{kg}{m^3}} = \frac{kg \times m^3}{ms \times kg} = m^2 s$

S.I Unit - $m^2 s$

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Stoke's Law:-

Spherical body.



Viscous force acting on the body due to surrounding fluid is,

$$F_v = 6\pi\eta r v$$

η = Coefficient of viscocity of surrounding fluid

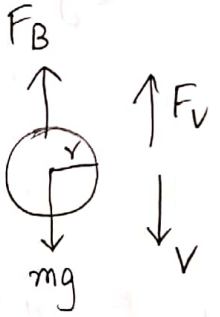
r = radius of body

V = Speed of body in fluid.

* Consider a sphere,

r = radius of sphere

ρ = density of sphere



Net force on the sphere,

$$F_{net} = mg - (F_B + F_v)$$

$$ma = mg - (F_B + F_v)$$

$$a = \frac{mg - (F_B + F_v)}{m}$$

$$\therefore a = \frac{dv}{dt}$$

Here $V \uparrow \Rightarrow F_v = 6\pi\eta r v \ (\uparrow)$

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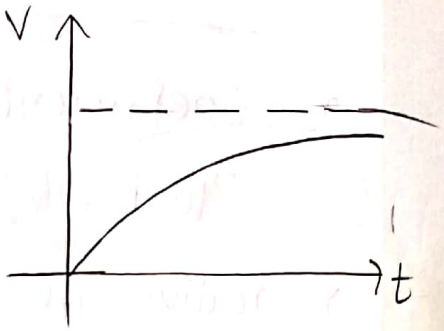
$\Rightarrow F_B + F_V$ increases

$\therefore$ acceleration $\rightarrow$ decreases

When sphere attains "terminal velocity (v_t)"

↳ Maximum Velocity

$$\frac{dv}{dt} = 0 \Rightarrow a = 0$$



$$\Rightarrow mg - (F_B + F_V) = 0$$

$$\boxed{mg = F_B + F_V}$$

$$\frac{4}{3}\pi r^3 \rho g = \frac{4}{3}\pi r^3 \sigma g + 6\pi\eta r v_t.$$

σ = density of surrounding fluid.

$$\frac{2}{3}(\rho - \sigma) g r^3 = 3\eta r v_t$$

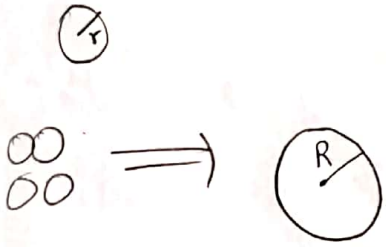
$$\boxed{v_t = \frac{2r^2}{9\eta}(\rho - \sigma) g}$$

From this, $v_t \propto r^2$

$$v_t \propto \frac{(\rho - \sigma)}{\eta}$$

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Rain drops:-



N-drops merged.

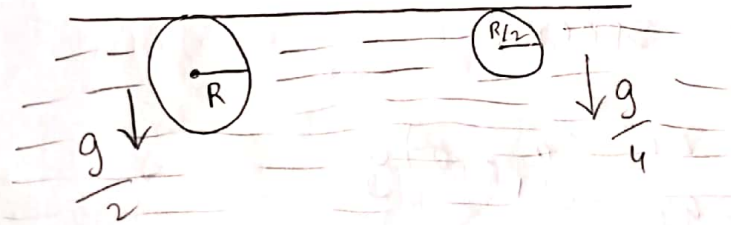
$$N\frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$$

$$Nr^3 = R^3.$$

$$R^2 = N^{2/3} r^2$$

$$\boxed{V_{big} = N^{2/3} V_{small}}$$

Ex:- Two balls of radii R & $\frac{R}{2}$ are released inside a deep water tank. Their initial accelerations are found to be $\frac{g}{2}$ & $\frac{g}{4}$ respectively. Find the velocity of smaller ball relative to the larger ball a long time after two balls are released.



Sol:- Initially, v=0

$mg - F_B = ma$ [For big sphere]

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$$\frac{4}{3}\pi r^3 \rho_1 g - \frac{4}{3}\pi r^3 \sigma g = \frac{4}{3}\pi r^3 \rho_1 \left(\frac{g}{2}\right) \rightarrow (1)$$

$$\rho_1 - \sigma = \frac{\rho_1}{2}$$

$$\boxed{\sigma = \frac{\rho_1}{2}} \rightarrow (1)$$

$mg - F_B = ma$ [For small sphere].

$$\frac{4}{3}\pi \left(\frac{r}{2}\right)^3 \rho_2 g - \frac{4}{3}\pi \left(\frac{r}{2}\right)^3 \sigma g = \frac{4}{3}\pi \left(\frac{r}{2}\right)^3 \rho_2 \left(\frac{g}{4}\right)$$

$$-\sigma + \rho_2 = \frac{\rho_2}{4}$$

$$\frac{3\rho_2}{4} = \sigma$$

$$\boxed{\rho_2 = \frac{4\sigma}{3}} \rightarrow (2)$$

$$\frac{\rho_1}{2} = \frac{3\rho_2}{4} \Rightarrow \boxed{\rho_1 = \frac{3}{2}\rho_2}$$

Terminal Velocity,

For big sphere,

$$V_1 = \frac{2r^2}{9\eta}(\rho_1 - \sigma)g.$$

$$V_1 = \frac{2r^2}{9\eta}(2\sigma - \sigma)g$$

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$$v_1 = \frac{2r^2\sigma g}{9\eta} \quad (\downarrow)$$

For small sphere,

$$v_2 = \frac{2}{9\eta}\left(\frac{r}{2}\right)^2\left(\frac{4}{3}\sigma - \sigma\right)g.$$

$$v_2 = \frac{r^2}{18\eta}\left(\frac{\sigma}{3}\right)g \quad (\downarrow)$$

Relative Velocity,

$$v_{rel} = v_2 - v_1$$

$$v_{rel} = \frac{r^2\sigma g}{\eta}\left(\frac{1}{54} - \frac{2}{9}\right)$$

$$v_{rel} = \frac{-11}{54}\frac{r^2\sigma g}{\eta} \quad (\downarrow)$$

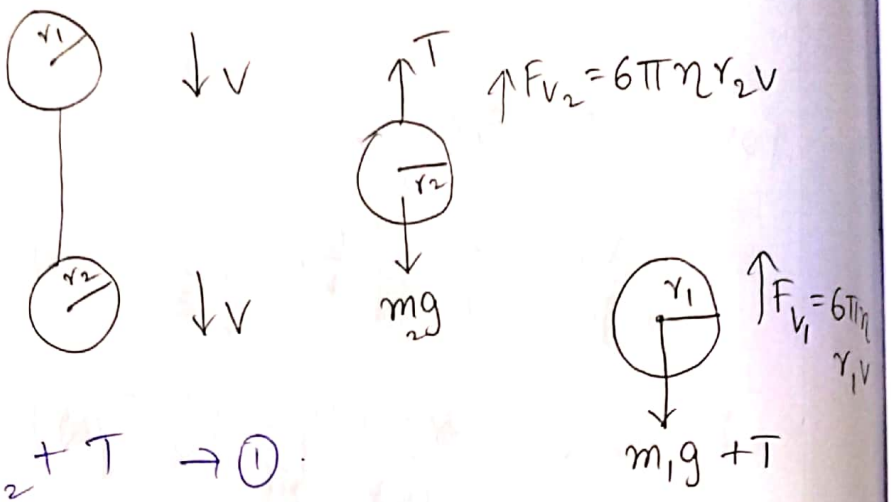
$$\boxed{v_{rel} = \frac{11r^2\sigma g}{54\eta} \quad (\uparrow)}$$

Ex:- Two balls of same material of density ρ but radius R_1 & R_2 are joined by a light inextensible vertical thread and released from large height in a medium of coefficient of viscocity η. Find the terminal velocity acquired by balls. Also find tension in string connecting both the balls when both are moving with terminal

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Velocity. Neglect buoyancy force.

Sol:-



$m_2 g = F_{V_2} + T \rightarrow (1)$

$m_1 g = F_{V_1} - T \rightarrow (2)$

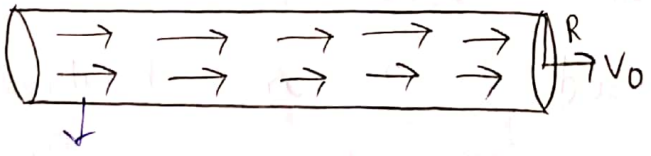
(1) + (2)

$$m_1 g + m_2 g = F_{V_1} + F_{V_2}$$

$$\frac{4}{3}\pi g (r_1^3 + r_2^3) = 6\pi\eta v (r_1 + r_2)$$

$$v = \frac{2\rho g (r_1^3 + r_2^3)}{9\eta (r_1 + r_2)}$$

Poiseulle's Equation :-



Ideal fluid

* Velocity of liquid is same at all points

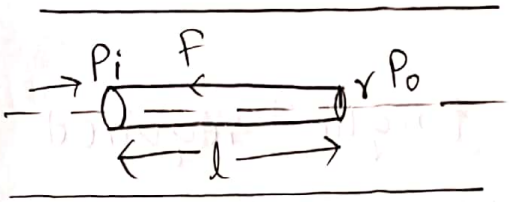
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Rate of flow of liquid through tube

$$Q = AV_0 = \pi R^2 V_0$$

Real liquid flowing through Capillary tube ① Velocity of fluid at walls = 0

② Maximum at axis of tube



$$f = \eta A \cdot \frac{dv}{dr} = \eta \cdot 2\pi r l \frac{dv}{dr}$$

Force due to pressure difference,

$$F = (\Delta P)\pi r^2$$

For Steady flow,

$$(\Delta P)\pi r^2 + \eta (2\pi r l)\frac{dv}{dr} = 0$$

$$\Delta P \pi r^2 = -2\eta r l \frac{dv}{dr}$$

$$\int_{r=r}^{r=R} \frac{\Delta P}{2\eta l} r \cdot dr = -\int_{v}^{0} dv$$

$$\frac{\Delta P}{2\eta l}\left(\frac{R^2 - r^2}{2}\right) = V$$

$$V = \frac{\Delta P}{4\eta l}(R^2 - r^2)$$

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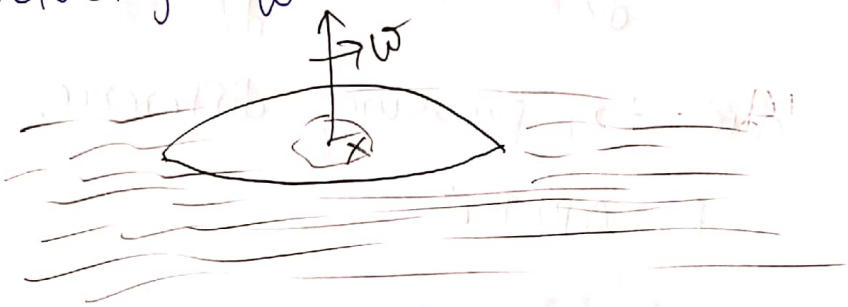
⇒ Velocity Profile (Parabola)

At the axis of tube, $r=0$

$$V_{max} = \frac{\Delta P R^2}{4\eta l}$$

$\Delta P = P_i - P_o$ = Pressure difference across the ends of tube

Ex:- Find the torque required to rotate the disc with constant angular velocity ω.



Sol:- $dA = 2\pi x\, dx$

$$dF = \eta (dA)\left(\frac{dv}{dy}\right)$$

$$dF = \eta_0 x \left(2\pi x\, dx\right)\left(\frac{x\omega - 0}{T}\right)$$

$$d\tau = (dF)x$$

$$= \frac{2\pi \eta_0 \omega x^4 dx}{T}$$

$$\tau = \int_{x=0}^{x=R} d\tau = \frac{2\pi \eta_0 \omega R^5}{5T}$$

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Ex:- A spherical ball of radius r is dropped in a viscous liquid having density ρ, coefficient of viscocity η.

Calculate (1) Power of gravitational force acting on ball at a time t after it is dropped (P_1)

(2) rate of heat generation due to rubbing of fluid molecules with the ball at time after it is dropped.

(3) How do P_1 & P_2 change if radius of ball is doubled.

(4) Find P_1 & P_2 when both become equal

Sol:- At any time 't',

Net force acting on the body,

$$F_{net} = mg - F_B - F_V$$

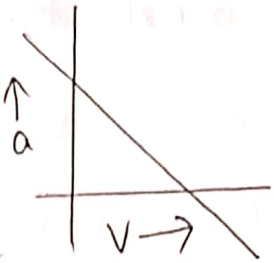
$$ma = mg - F_B - F_V$$

$$\frac{4}{3}\pi r^3 d\left(\frac{dv}{dt}\right) = \frac{4}{3}\pi r^3 dg - \frac{4}{3}\pi r^3 \rho g - 6\pi\eta r v.$$

$$\frac{dv}{dt} = g\left(1 - \frac{\rho}{d}\right) - \frac{9\eta v}{2r^2 d}$$

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$$\frac{dv}{dt} = a - bv$$



$$a = g\left(1 - \frac{\rho}{d}\right)$$

$$b = \frac{9\eta}{2r^2 d}$$

$$\int_0^v \frac{dv}{a - bv} = \int_0^t dt$$

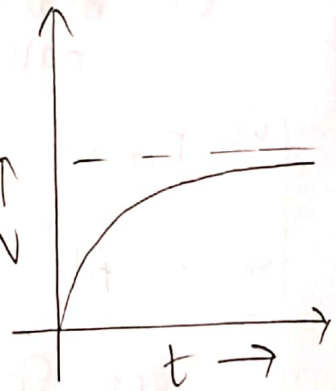
$$\frac{\left[\ln(a - bv)\right]_0^v}{-b} = t$$

$$\frac{-1}{b}\left[\ln(a - bv) - \ln a\right] = t$$

$$\ln\left(\frac{a - bv}{a}\right) = -bt$$

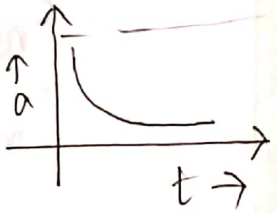
$$\frac{a - bv}{a} = e^{-bt}$$

$$1 - \frac{bv}{a} = e^{-bt}$$



$$\frac{bv}{a} = 1 - e^{-bt}$$

$$v = \frac{a}{b}\left(1 - e^{-bt}\right)$$



$$v = \frac{g\left(1 - \frac{\rho}{d}\right)2r^2 d}{9\eta}\left(1 - e^{\frac{-9\eta}{2r^2 d}t}\right)$$

$$\boxed{v = \frac{2r^2 g}{9\eta}(d - \rho)\left(1 - e^{\frac{-9\eta t}{2r^2 d}}\right)}$$

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At $t = \alpha$

$$V = \frac{a}{b} \text{ (max)}$$

$$\boxed{V_t = \frac{2r^2 g}{9\eta}(d-\rho)}$$

① Power developed due to gravity.

$$P_1 = mgv \qquad [P_{inst} = \vec{F}\cdot\vec{v}]$$

$$P_1 = \frac{4}{3}\pi r^3 dg\left[\frac{g\left(1-\frac{\rho}{d}\right)2r^2 d}{9\eta}\left(1-e^{\frac{-9\eta t}{2r^2 d}}\right)\right]$$

$$P_1 = \frac{8\pi r^5 dg^2(d-\rho)}{27\eta}\left(1-e^{\frac{-9\eta t}{2r^2 d}}\right)$$

Clearly, $\boxed{P_1 \propto r^5}$

② Power dissipate due to viscous force

$$P_2 = (F_v)v$$

$$P_2 = (6\pi\eta r v)v$$

$$P_2 = 6\pi\eta r v^2$$

$$P_2 = 6\pi r \times \frac{4r^4}{81\eta^2} \times \eta \times g^2 (d-\rho)^2 \left(1-e^{\frac{-9\eta}{2r^2 d}}\right)^2$$

Clearly $P_2 \propto r^5$

$K.E = \frac{1}{2}mv^2$

$K.E \propto r^3 \cdot (r^2)^2$

$\boxed{K.E \propto r^7}$

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(3) If radius is doubled,

P_1 & P_2 will be 32 times the previous one.

(4) For $d >> \ell$, $\Rightarrow d - \ell \approx d$

At $t = \infty$ we have, $e^{-\infty} = 0$

$$P_1 = P_2$$

$$P_1 = P_2 = \frac{8\pi r^5 d^2 g^2}{27\eta}$$

Poiseulle's Equation:-

Velocity Profile,

$$v = \frac{\Delta P}{4\eta \ell}(R^2 - r^2)$$

At $r = 0 \Rightarrow V_0 = \frac{\Delta P R^2}{4\eta \ell} = V_{max}$

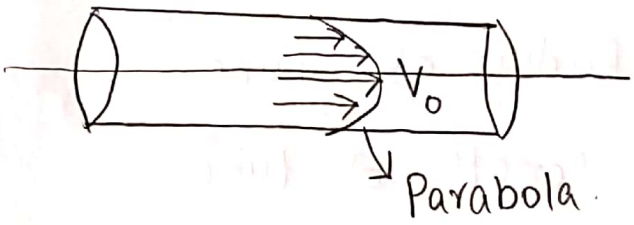
$$\boxed{v = V_{max}\left(1 - \frac{r^2}{R^2}\right)}$$

Velocity gradient,

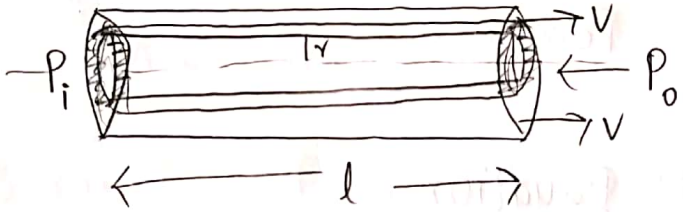
$$\frac{dv}{dr} = \frac{-2V_{max}\, r}{R^2}$$

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-ve sign indicates as 'r' increases from axis of tube, 'v' decreases.



Rate of flow of liquid through capillary tube :-



Rate of flow through elementary shell,

$$dQ = (dA)v$$

$$= (2\pi r\, dr)v$$

$$= (2\pi r\, dr)\frac{\Delta P}{4\eta l}(R^2 - r^2)$$

$$dQ = \frac{\pi \Delta P}{2\eta l}(R^2 r - r^3)\, dr$$

Total rate of flow through tube,

$$Q = \int_{r=0}^{r=R} dQ = \frac{\pi \Delta P}{2\eta l}\int_{r=0}^{r=R} R^2 r - r^3 \cdot dr$$

$$\boxed{Q = \frac{\pi \Delta P R^4}{8\eta l}} \rightarrow \text{Poiseuille's Equation}$$

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where $\Delta P = P_i - P_o$ = Pressure difference across the ends of tube

R = Radius of tube.

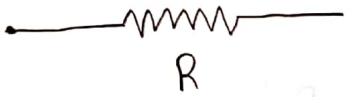
l = length of tube

η = Coefficient of Viscocity of liquid flowing through tube

Important Points:-

(1) Poiseullie's Equation,

$$Q = \frac{\pi(\Delta P)R^4}{8\eta l} = \frac{\Delta P}{\left(\frac{8\eta l}{\pi R^4}\right)} = \frac{\Delta P}{R_F}$$

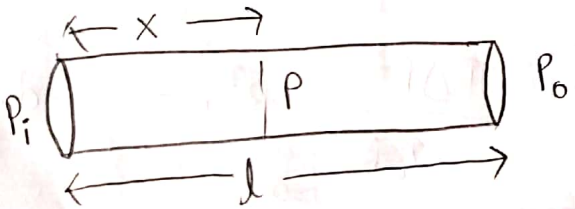


From Ohm's Law, $\Delta V = IR$

$$I = \frac{\Delta V}{R}$$

Similarly, $Q = \frac{\Delta P}{R_F}$

(2)



P = Pressure in the fluid at a distance 'x'

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Here $Q_1 = Q_2$

$$\frac{\pi (P_i - P) R^4}{8\eta x} = \frac{\pi (P - P_0) R^4}{8\eta (l - x)}$$

$$\frac{P_i - P}{x} = \frac{P - P_0}{l - x}$$

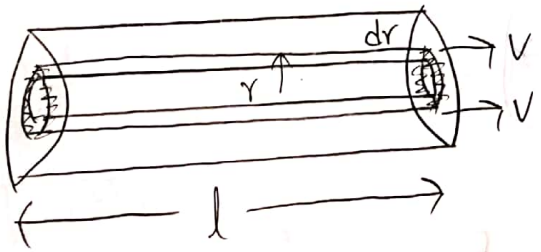
$$P_i (l - x) - P(l - x) = P_0 x - P_0 x$$

$$P_i (l - x) + P_0 x = P[x + l - x]$$

$$P = \frac{P_i l - (P_i - P_0) x}{l}$$

$$\boxed{P = P_i - \left(\frac{P_i - P_0}{l}\right) x}$$

Momentum of liquid flowing through Capillary Tube:-



$$dP = (dm) V$$

$$dP = (2\pi r\, dr)\, l\, \rho\, \frac{\Delta P}{4\eta l} (R^2 - r^2)$$

Total momentum, $P = \int_{r=0}^{r=R} dP$

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K.E of liquid flowing through tube :-

$$dK = \frac{1}{2} dm v^2$$

$$= \frac{1}{2} (2\pi r\, dr)\, l\, \rho \left[\frac{\Delta P}{4\eta l}(R^2 - r^2)\right]^2$$

Total K.E,

$$K.E = \int_{r=0}^{r=R} dK$$

Viscous force acting b/w layers of fluid

$$F = \eta A \left(\frac{dv}{dr}\right)$$

$$= \eta (2\pi r l) \frac{d}{dr}\left(\frac{\Delta P}{4\eta l}(R^2 - r^2)\right)$$

$$= \eta (2\pi r l) \frac{\Delta P}{4\eta l}(-2r)$$

$$= -\pi \Delta P r^2$$

$$\boxed{F_v \propto r^2}$$

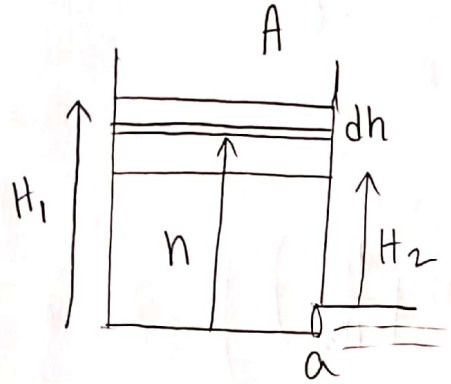
At walls,

$$F = \eta (2\pi R l) \frac{d}{dr}\left(\frac{\Delta P}{4\eta l}(R^2 - 2r)\right)$$

$$= 2\pi \eta R l \times \frac{\Delta P}{4\eta l}$$

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(1)



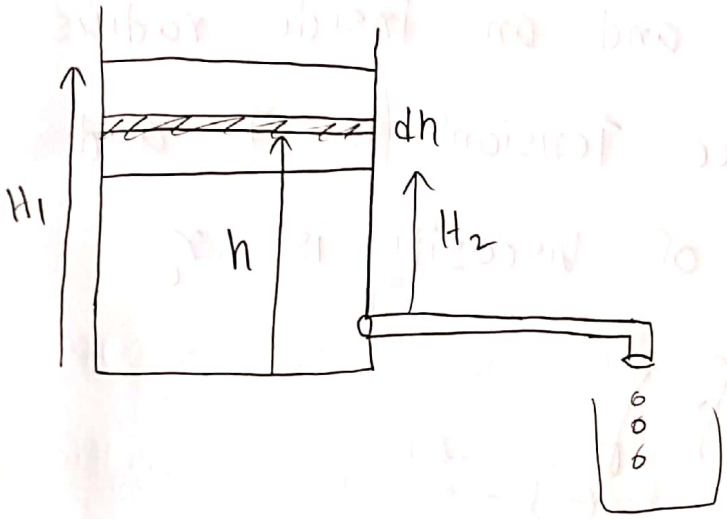
Time taken to empty the liquid from H_1 to H_2 is.

$$t = \sqrt{\frac{2}{g}} \frac{A}{a} \left(\sqrt{H_1} - \sqrt{H_2}\right)$$

$$-\frac{A\,dh}{dt} = aV$$

$$\hookrightarrow V = \sqrt{2gh}$$

(2)



When Capillary tube is taken,

$$-\frac{A\,dh}{dt} = \frac{\Delta P\, \pi R^4}{8\eta l}$$

$$\Delta P = P_i - P_o$$

$$= P_o + \rho g h - P_o$$

$$= \rho g h$$

$$-A\frac{dh}{dt} = \frac{\pi (\rho g h) R^4}{8\eta l}$$

$$\int_{h=H_1}^{h=H_2} \frac{-dh}{h} = \frac{\pi \rho g R^4}{8\eta l A} \int_0^t dt$$

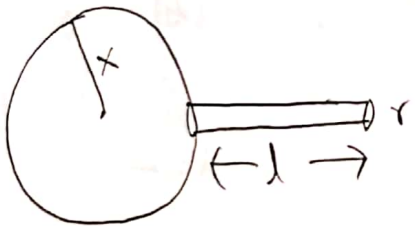
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$$\ln\left(\frac{H_1}{H_2}\right) = \frac{\pi \rho g R^4}{8\eta l A} \cdot t$$

$$t = \frac{8\eta l A}{\pi \rho g R^4} \ln\left(\frac{H_1}{H_2}\right)$$

Ex:- Find the lifetime of a soap bubble of radius R connected with the atmosphere through a capillary of length 'l' and an inside radius 'r'. The Surface Tension is T and Coefficient of Viscosity is η.

Sol:-



$$\frac{-dv}{dt} = \frac{\pi (\Delta P) r^4}{8\eta l}$$

$$\Delta P = P_0 + \frac{4T}{x} - P_0$$

$$= \frac{4T}{x}$$

$$v = \frac{4\pi x^3}{3}$$

$$\frac{-d}{dt}\left(\frac{4}{3}\pi x^3\right) = \frac{\pi \left(\frac{4T}{x}\right) r^4}{8\eta l}$$

$$\frac{-4}{3}\pi \cdot 3x^2 \cdot \frac{dx}{dt} = \frac{\pi \cdot 4T r^4}{8\eta x l}$$

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$$\int_{R}^{0} \frac{-8x^3 \eta l}{\pi r^4} \cdot dx = \int_{0}^{t} dt$$

$$t = \frac{2\eta l R^4}{\pi r^4}$$

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